

CH4 Distribution Functions and Discrete Random Variables

4.1 Random variables

Def A real-valued function $X: S \rightarrow R$ is called a random variable of the experiment if, for each interval $I \subseteq R$, $\{s : X(s) \in I\}$ is an event.

In probability, the set $\{s : X(s) \in I\}$ is often abbreviated as $\{X \in I\}$ or simply as $X \in I$

4.2 Distribution functions

CDF: (**cumulative**) distribution function

$$F_x(t) = F(t) = P(X \leq t)$$

1. F is nondecreasing; that is, if $t < u$, then $F(t) \leq F(u)$.

$$2. \quad \lim_{t \rightarrow \infty} F(t) = 1 \quad \lim_{t \rightarrow -\infty} F(t) = 0$$

3. F is right continuous; that is, for every t in $\mathbb{R}$, $F(t^+) = F(t)$

Event concerning X	Probability in terms of F	Event concerning X	Probability in terms of F
$X \leq a$	$F(a)$	$a < X \leq b$	$F(b) - F(a)$
$X > a$	$1 - F(a)$	$a < X < b$	$F(b^-) - F(a)$
$X < a$	$F(a^-)$	$a \leq X \leq b$	$F(b) - F(a^-)$
$X \geq a$	$1 - F(a^-)$	$a \leq X < b$	$F(b^-) - F(a^-)$
$X = a$	$F(a) - F(a^-)$		

4.3 Discrete random variables

Discrete (離散的)

A set of values that is countable is called discrete

Countable: 有空隙在每個數值之間, 才可以數

Examples of set measure

- Finite: $\{0, 1, 2\}$
- Finite: $\{1, 2, 3, 4, \dots\}$
- Infinite: $\{x : x > 0\}$ (You can not find what is the next of zero, 0.1 0.01 ?)

P.F.M (Probability mass function)

- (a) $p(x) = 0$ if $x \notin \{x_1, x_2, x_3, \dots\}$
- (b) $p(x_i) = P(X = x_i)$ and hence $p(x_i) \geq 0$
- (c) $\sum_{i=1}^{\infty} p(x_i) = 1$

4.4 Expectations of discrete random variables

Expected value

$$E(X) = \sum_{x \in A} xp(x)$$

Called: mean, expectation of X

Denoted: EX , μ_x , μ

$$\begin{aligned} p30. \quad \sum_{k=n}^N \binom{k}{n} &= \binom{N+1}{n+1} \\ &= \binom{n}{n} + \binom{n+1}{n} + \binom{n+2}{n} + \dots + \binom{N}{n} \end{aligned}$$

期望值遵守線性規則（三條）

Var 沒有遵守線性三條

Theorem 4.5

$$Var(aX + b) = a^2 Var(X)$$

→ Var 在計算實際情與平均值的差異（距離）

